TSAR Project Part 1 – Exploratory Data Analysis

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```r
# # Random sample your group's data set from the original LCdata data set
# LCdata <- fread("LCdata.csv")
# set.seed(1)
# myLCdata <- LCdata %>% sample(5) %>% slice_sample(prop = .5)
# fwrite(myLCdata, file = "myLCdata.csv")
myLCdata <- fread("myLCdata.csv")
```

1 Task 1: Business Understanding

```r
glimpse(myLCdata)
```

```
Rows: 44,369
Columns: 5
$ annual_inc     <dbl> 60000, 18576, 40000, 35000, 125000, 144000, 205000, 118~
$ open_acc       <int> 6, 7, 15, 3, 12, 8, 9, 18, 12, 3, 11, 10, 15, 5, 13, 10~
$ purpose        <chr> "credit_card", "debt_consolidation", "small_business", ~
$ home_ownership <chr> "OWN", "RENT", "RENT", "OWN", "RENT", "MORTGAGE", "MORT~
$ inq_last_6mths <int> 0, 0, 2, 1, 0, 2, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 1, 3, 0~
```

```r
str(myLCdata)
```

```
Classes 'data.table' and 'data.frame':  44369 obs. of  5 variables:
 $ annual_inc    : num  60000 18576 40000 35000 125000 ...
 $ open_acc      : int  6 7 15 3 12 8 9 18 12 3 ...
 $ purpose       : chr  "credit_card" "debt_consolidation" "small_business" "home_improvemen
```

```
$ home_ownership: chr  "OWN" "RENT" "RENT" "OWN" ...
$ inq_last_6mths: int  0 0 2 1 0 2 0 0 1 0 ...
- attr(*, ".internal.selfref")=<externalptr>
```

1.1 Variable Overview and Business Interpretation

The dataset `myLCdata` contains 44,369 observations and 5 variables. Based on the output of `glimpse()` and `str()`, the variables can be categorized as follows:

1.1.1 Variable Description

- **annual_inc (numeric)**
  The borrower's self-reported annual income in US dollars. This variable is continuous and measures repayment capacity.
- **open_acc (integer)**
  The number of open credit accounts the borrower currently has. This reflects the borrower's existing credit exposure.
- **purpose (character)**
  The stated purpose of the loan (e.g., credit card refinancing, debt consolidation, small business, home improvement). This is a categorical variable describing how the borrower intends to use the funds.
- **home_ownership (character)**
  The borrower's home ownership status (e.g., OWN, RENT, MORTGAGE). This variable provides information about the borrower's financial stability and asset ownership.
- **inq_last_6mths (integer)**
  The number of credit inquiries made in the last six months. This is an indicator of recent credit-seeking behavior and potential financial stress.

1.2 Business Context in Peer-to-Peer Lending

In a peer-to-peer lending platform such as Lending Club, investors directly fund loans to borrowers. The platform evaluates borrower characteristics to estimate credit risk.

The variable **annual_inc** is central for assessing repayment ability, as higher income generally implies a greater capacity to service debt.

The variable **open_acc** reflects the borrower's existing financial obligations. A high number of open accounts may indicate greater financial leverage and potentially higher risk.

The variable **purpose** helps categorize the type of loan demand. For example, debt consolidation loans may indicate existing debt burdens, whereas home improvement loans may reflect investment in property.

The variable **home_ownership** can serve as a proxy for financial stability. Borrowers who own property or have a mortgage may be perceived as more financially stable compared to renters.

Finally, **inq_last_6mths** measures recent credit inquiries. A high number of inquiries can signal increased borrowing activity and may be associated with higher default risk.

Overall, the selected variables capture borrower income, credit exposure, loan motivation, housing stability, and recent credit behavior — all of which are relevant for evaluating creditworthiness in a peer-to-peer lending context.

2 Task 2: Data Understanding Based on Summary Statistics

2.1 Summary Statistics

```r
summary(myLCdata)
```

```
   annual_inc         open_acc       purpose          home_ownership
 Min.   :   4000   Min.   : 1.00   Length:44369       Length:44369
 1st Qu.:  45649   1st Qu.: 8.00   Class :character   Class :character
 Median :  65000   Median :11.00   Mode  :character   Mode  :character
 Mean   :  75269   Mean   :11.57
 3rd Qu.:  90000   3rd Qu.:14.00
 Max.   :6000000   Max.   :76.00
                   NA's   :2
 inq_last_6mths
 Min.   : 0.0000
 1st Qu.: 0.0000
 Median : 0.0000
 Mean   : 0.6966
 3rd Qu.: 1.0000
 Max.   :31.0000
 NA's   :2
```

```r
# install skimr if necessary
if (!require("skimr")) install.packages("skimr")
library(skimr)

skim(myLCdata)
```

Table 1: Data summary

Name	myLCdata
Number of rows	44369
Number of columns	5
Key	NULL
Column type frequency:	
character	2
numeric	3
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
purpose	0	1	3	18	0	14	0
home_ownership	0	1	3	8	0	6	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
annual_inc	0	1	75269.01	60410.50	4000	45649	65000	90000	6000000	
open_acc	2	1	11.57	5.38	1	8	11	14	76	
inq_last_6mths	2	1	0.70	1.01	0	0	0	1	31	

2.2 Interpretation and Domain Context

2.2.1 1. Missing Values (NAs)

- Are missing values present?

- In which variables?
- How many?
- Is this plausible from a business perspective?
- Would this require data preparation?

Discussion: > Write interpretation here.

2.2.2 2. Numerical Variables

For each numeric/integer variable:

2.2.2.1 Variable: _____

- Mean:
- Median:
- Minimum:
- Maximum:
- Standard deviation:
- Is the distribution symmetric or skewed?
- Are there potential outliers?
- Do the extreme values seem realistic in the Lending Club context?

Statistical Interpretation: > Write here.

Domain Interpretation: > Write here.

2.2.3 3. Categorical Variables

For each categorical variable:

2.2.3.1 Variable: _____

- Number of categories:
- Most frequent category:
- Least frequent category:
- Are categories balanced or highly unbalanced?
- Does this distribution make business sense?

Statistical Interpretation: > Write here.

Domain Interpretation: > Write here.

2.2.4 4. Key Insights from Summary Statistics

- Most important statistical findings:
- Potential data quality issues:
- First hypotheses about borrower behavior:

---

3 Task 3: Univariate Exploratory Data Analysis (EDA) with ggplot2

3.1 Variable 1: _____

3.1.1 Visualization

```
# Insert ggplot code here
```

3.1.2 Why This Plot Type?

- Why is this diagram appropriate for the variable type?
- What can be read from this diagram type (e.g., range, distribution, frequency)?

3.1.3 Interpretation

- What does the distribution look like?
- Are there outliers?
- Is the distribution skewed?
- What does this imply in the business context?
- Does this confirm findings from Task 2?

3.2 Variable 2: _____

3.2.1 Visualization

```
# Insert ggplot code here
```

3.2.2 Interpretation

Write here.

3.3 Variable 3: _____

3.3.1 Visualization

```
# Insert ggplot code here
```

3.3.2 Interpretation

Write here.

3.4 Variable 4: _____

3.4.1 Visualization

```
# Insert ggplot code here
```

3.4.2 Interpretation

Write here.

3.5 Variable 5: _____

3.5.1 Visualization

```
# Insert ggplot code here
```

3.5.2 Interpretation

Write here.

3.6 Overall Observations from Univariate EDA

- Which variables show skewness?
- Which variables show imbalance?
- Any surprising patterns?
- Do visual findings confirm summary statistics?

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4 Task 4: Bivariate Exploratory Data Analysis (EDA)

4.1 Pairs Plot Overview

```r
if (!require("GGally")) install.packages("GGally")
library(GGally)

ggpairs(myLCdata)
```

```
Warning: Removed 2 rows containing missing values
Removed 2 rows containing missing values

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_density()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).
Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

Warning: Removed 2 rows containing missing values

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

```
Warning: Removed 2 rows containing non-finite outside the scale range (`stat_bin()`).
Removed 2 rows containing non-finite outside the scale range
(`stat_boxplot()`).

Warning: Removed 2 rows containing missing values or values outside the scale range
(`geom_point()`).
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`stat_bin()` using `bins = 30`. Pick better value `binwidth`.

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

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Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_bin()`).

Warning: Removed 2 rows containing non-finite outside the scale range
(`stat_density()`).
```

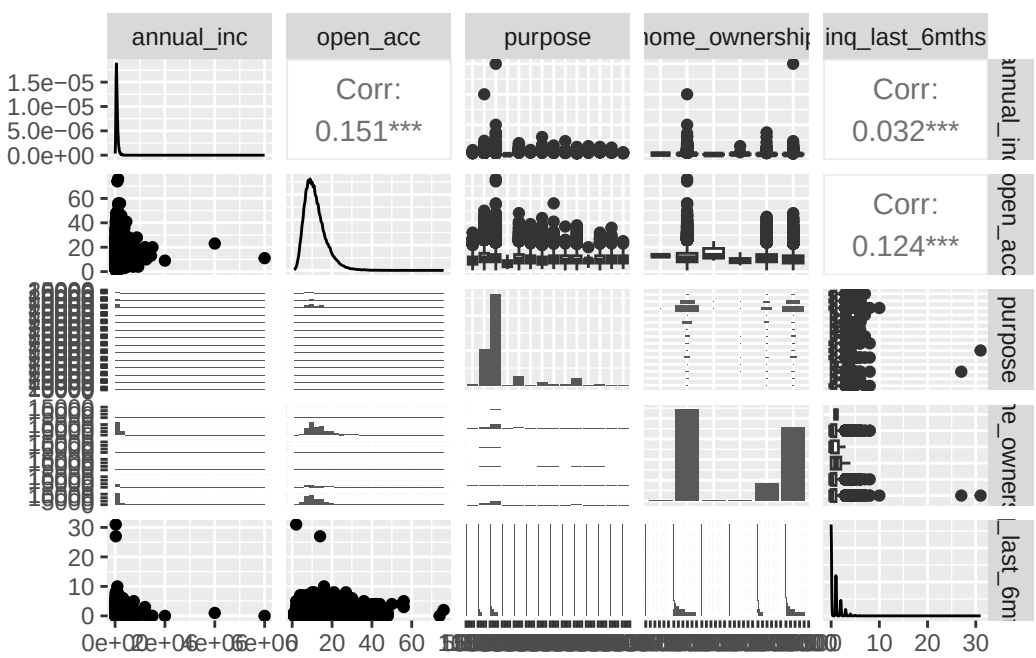

4.1.1 Initial Observations

- Are there visible correlations?
- Any clear patterns?
- Any surprising relationships?

4.2 Selected Insight 1

4.2.1 Variable Pair: _____ vs _____

4.2.2 Why This Pair Is Interesting

- What pattern was observed?
- Why was it not obvious before?

4.2.3 Stand-alone Professional Plot

```
# Recreate plot using ggplot2
```

4.2.4 Interpretation

- What does this relationship indicate?
- Business interpretation?

4.3 Selected Insight 2

4.3.1 Variable Pair: _____ vs _____

4.3.2 Stand-alone Professional Plot

```
# Recreate plot using ggplot2
```

4.3.3 Interpretation

Write here.

4.4 Adding a Third Variable

4.4.1 Enhanced Plot with Third Variable (e.g., color, shape, facet)

```
# ggplot with third aesthetic mapping
```

4.4.2 Why This Third Variable?

- Why is it relevant?
- What additional insight does it provide?
- What was not visible before?

4.5 Final Bivariate Insights

- Most important relationships discovered:
- Implications for credit risk assessment:
- Potential hypotheses for predictive modeling: